

Equivalence b/w ERM & distribution Est.

Suppose

$$D = \{x_i, y_i\}_{i=1}^n \sim \text{iid } P_{xy}$$

In the prob. approach, the main problem is to estimate distributions from D ,

$$\hookrightarrow P_x, P_{xy}, P_{y|x}$$

via divergence minimization.

In the risk minimization approach, the main problem is to estimate a function $h: X \rightarrow Y$, from D , via Empirical risk minimization.

prob. approach, Given D ,

$$\theta^* = \underset{\theta}{\operatorname{argmin}} D_{KL}(P_x \parallel P_\theta)$$

ERM, Given D ,

$$h^* = \underset{h \in \mathcal{H}}{\operatorname{argmin}} \frac{1}{N} \sum_{i=1}^N L(h(x_i), y_i)$$

claim: Div. minimization & ERM are equivalent

eg: suppose we want to estimate $p_{y|x}$
 $p_{\theta}(y|x) \sim N(y; h_{\theta}(x), I)$: Model

Max. likelihood objective:

$$\theta^* = \underset{\theta}{\operatorname{argmin}} \left[\frac{1}{N} \sum_{i=1}^n \|y_i - h_{\theta}(x_i)\|_2^2 \right]$$

() ERM with squared Error loss.

Linear Family of Models.

$$h_{\theta}(x) = \theta^T x + \theta_0 \quad \begin{array}{l} x \in \mathbb{R}^d, y \in \mathbb{R} \\ \theta \in \mathbb{R}^d, \theta_0 \in \mathbb{R} \end{array}$$

$$\text{with } x = [x \ 1]^T, \quad h_{\theta}(x) = \theta^T x, \quad \theta \in \mathbb{R}^{d+1}$$

ERM with Linear Models & sq. Error loss.

$$\begin{aligned}
 \theta^* &= \underset{\theta}{\operatorname{argmin}} \frac{1}{N} \sum_{i=1}^N \|h_{\theta}(x_i) - y_i\|_2^2 \\
 &= \underset{\theta}{\operatorname{argmin}} \frac{1}{N} \sum_{i=1}^N \|\theta^T x_i - y_i\|_2^2 \\
 &= (X^T X)^{-1} X^T y
 \end{aligned}$$

where $X = \begin{bmatrix} x_1^T \\ x_2^T \\ \vdots \\ x_n^T \end{bmatrix}$

$$y = [y_1, y_2, \dots, y_n]^T$$

Generalized Linear Models.

In the naive linear case,
 $h_{\theta}(x) = \theta_0 x^0 + \theta_1 x^1 + \dots + \theta_d x^d$

where x^j is the j^{th} data dimension

Instead of linear combinations, one can consider non-linear combinations of data dimensions.

$$h_0(x) = \theta_1(x^1)^3 + \theta_2(x^2)^4 + \dots$$

The above case can be represented via pre-computed transformations.

Given x , defined $\hat{x} = \phi(x)$ where ϕ is a fixed transformation.

$$\hat{x} = \left[(x^1)^2 \quad (x^2)^3 \quad (x^1)^2 + (x^2)^6 \quad \dots \right]$$

$$\phi(x) = \left[\phi_1(x^1) \quad \phi_2(x^1) \dots \phi_p(\text{data dim}) \right]$$

$$\begin{array}{ccc} \phi : X & \rightarrow & \hat{X} \\ \mathbb{R}^d \hookrightarrow & & \hookrightarrow \mathbb{R}^p \end{array}$$

ϕ : user defined design choice.

Subsequently solve ERM with a model linear in $\phi(x)$ space.

Given D , compute transformed data

$$\hat{D} = \left\{ \phi(x_i), y_i \right\}_{i=1}^N \quad \begin{array}{l} \phi(x_i) \in \mathbb{R}^p \\ y_i \in \mathbb{R} \end{array}$$

$$\hat{\theta}^* = \operatorname{argmin}_{\hat{\theta}} \frac{1}{N} \sum_{i=1}^N \left\| (\hat{\theta})^T \phi(x_i) - y_i \right\|_2^2$$

$$= \left(\Phi^T \Phi \right)^{-1} \Phi^T y$$

where $\Phi = \begin{bmatrix} \phi(x_1)^T \\ \phi(x_2)^T \\ \vdots \\ \phi(x_n)^T \end{bmatrix}$

eg: $x \in \mathbb{R}, y \in \mathbb{R}$

$$\phi(x) = \begin{bmatrix} 1 & x & x^2 \end{bmatrix}$$

ERM with $\theta^T \phi(x)$ is same as
ERM with $a_1 x + a_2 x^2 + 1$.

Linear Models for Classification.

$$\mathcal{D} = \left\{ x_i, y_i \right\}_{i=1}^n \sim \text{iid } \mathbb{P}_{x,y}$$

$$x_i \in \mathbb{R}^d, \quad y_i \in \{0, 1, 2, \dots, K\}$$

$$h_\theta(x) : \mathcal{X} \rightarrow \mathcal{Y}$$

$h_0(x) = \theta^T x$, with this the range space of $h_0(x)$ won't be $\mathcal{Y}$.

$$h_0(x) = \sigma(\theta^T x)$$

where $\sigma(t) = \frac{1}{1 + e^{-t}} \in [0, 1]$

sigmoid function.

final classifier

$$h(x) = \begin{cases} 1 & \text{if } h_0(x) > \tau \\ 0 & \text{otherwise} \end{cases}$$

For k -class classification.

$$x \in \mathbb{R}^d$$

$$y \in \{0, 1, 2, \dots, k-1\}$$

define $h_0(x) = \theta \underset{k \times d}{X} \underset{d \times 1}{\begin{matrix} \\ \end{matrix}}$

$$\theta \in \mathbb{R}^{k \times d}$$

objective : Transform a k -dimensional real vector into k -dim one-hot.

suppose $v \in \mathbb{R}^k$

define a function $\text{softmax}(v)$

$$\text{softmax}(v)_j = \frac{\exp(v_j)}{\sum_j \exp(v_j)}$$

$$z_j = \text{softmax}(v)_j,$$

$$\text{then } z = [z_1, z_2, \dots, z_k]$$

$$0 \leq z_j \leq 1, \quad \sum_j z_j = 1.$$

In the context of linear-models,
for k -class classification

$$h_\theta(x) = \text{softmax}(\theta_{k \times d} X_{d \times 1})$$

The range of $h_\theta(x)$ is $[0, 1]^k$

Logistic classifier / Regressors

ERM with logistic regression.

$$\theta^* = \operatorname{argmin}_{\theta} \frac{1}{N} \sum_{i=1}^N \left\| \operatorname{softmax}(\theta x) - y \right\|_2^2$$

The above optimization problem can't be solved analytically.

Solve it numerically, typically using 1st order grad. descent.

Gradient descent Method.

Suppose there is a function $\hat{R}(\theta)$ which needs to be optimized.

$\hat{R}(\theta)$ is a scalar function of parameters $\theta \in \mathbb{R}^P$

Grad. descent does it iteratively

Initialize θ^0

Tell convergence do:

$$\theta^{t+1} \leftarrow \theta^t - \alpha \nabla_{\theta} \hat{R}(\theta), \quad \nabla_{\theta} \hat{R}(\theta) \in \mathbb{R}^p$$

Cross-Entropy loss

$$h_{\theta}(x), y \in [0, 1]^k$$

$$L_c(h_{\theta}(x), y) = - \sum_k y_k \log h_{\theta}(x)$$